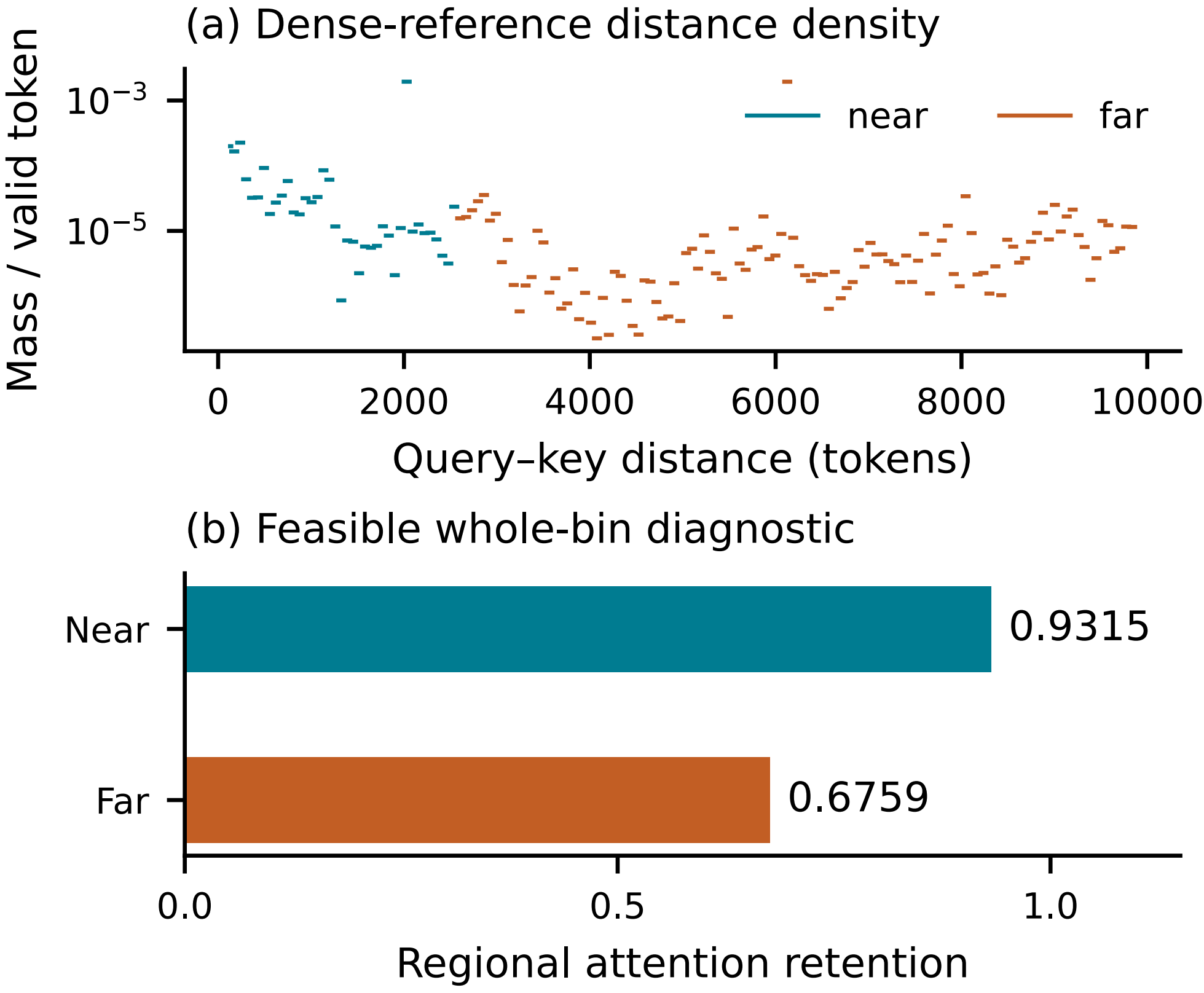

(a) Dense-reference distance density
Mass / valid token
$10^{-3}$
$10^{-5}$
near
far
0
2000
4000
6000
8000
10000
Query–key distance (tokens)
(b) Feasible whole-bin diagnostic
Near
Far
0.9315
0.6759
0.0
0.5
1.0
Regional attention retention